UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 003 · Particular integral Type 1 ($X = e^{ax}$) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·8	Unit 2 deck 003 (“Particular integral of Type 1”), slide 8 (rendered .pptx)
T003 00:10:15	Unit 2 AV-summary transcript 003 (“Particular integral of standard functions – Type 1”, 21 min), segment start
T004a 00:02:06	Unit 2 AV-summary transcript in bucket 004 (“Particular integral Type 1 and Type 2”, 12.6 min); only its Type 1 part (00:00:00–00:06:04) and the constant term of its $\sin^2 x$ example are cited here. The class 004 pack cites the same file.
CW U2 P10	Unit 2 class-work problem 10
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

Contents

1. Part A · Formula sheet
2. 1 · The inverse differential operator and the particular integral
3. 2 · Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$
4. 3 · Type 1: $X = e^{ax}$ and the cases of failure
5. 4 · Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x
6. Part B · Symbol glossary

A Formula sheet

1 The inverse differential operator and the particular integral

E1.1 · Inverse operators

ESA ★★

$$f(D)\left\{\frac{1}{f(D)}X\right\} = X$$

GIVES The defining property of $\frac{1}{f(D)}$.

APPLIES Any linear operator $f(D)$ with constant coefficients.

NOTE The order can be swapped: $\frac{1}{f(D)}\{f(D)X\} = X$ (T003 00:02:40), up to the CF.

S003·4 · T003 00:02:14 · T003 00:02:40

E1.2 · General solution of $f(D)y = X$

ESA ★★★

$$y = y_c + y_p = \text{CF} + \frac{1}{f(D)}X$$

GIVES The complete solution.

APPLIES $X \neq 0$; the CF from the auxiliary equation (class 002).

NOTE CF has n arbitrary constants; the PI has none.

S003·6 · S003·12 · T003 00:05:32 · T003 00:06:07

2 Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$

E2.1 · $\frac{1}{D}$ is integration

ESA ★★

$$\frac{1}{D}X = \int X\,dx, \qquad \frac{1}{D^2}X = \iint X\,dx\,dx$$

GIVES $\frac{1}{f(D)}X$ for $f(D) = D$ or D^2.

APPLIES X any function of x.

NOTE No constant of integration: $\frac{1}{D}x^2 = \frac{x^3}{3}$.

S003·7 · T003 00:06:24 · T003 00:06:40 · T003 00:06:50

E2.2 · $\frac{1}{D-a}$ and $\frac{1}{D+a}$ ESA ★★

$$\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx, \qquad \frac{1}{D+a}X = e^{-ax}\int e^{ax}X\,dx$$

GIVES The PI for one linear factor and any X.

APPLIES a a constant.

NOTE "This is the formula you should remember" (T003 00:07:29). The $D+a$ form: replace a by $-a$ (lecture only). Used for e^{e^x} in class 009 (ESA Dec-23 3c).

S003·7 · T003 00:07:21 · T003 00:07:29 · T003 00:07:45 · ESA Dec-23 3c

E2.3 · Product of linear factors ESA ★

$$\frac{1}{(D-a)(D-b)}X = \frac{1}{D-a}\left(\frac{1}{D-b}X\right)$$

GIVES Apply the factors one at a time (innermost first).

APPLIES $f(D) = (D-a)(D-b)$.

NOTE Or by partial fractions: $\frac{1}{a-b}\left[\frac{1}{D-a} - \frac{1}{D-b}\right]X$ (T003 00:09:02).

S003·7 · T003 00:08:07 · T003 00:08:25 · T003 00:09:02

3 Type 1: $X = e^{ax}$ and the cases of failure

E3.1 · Type 1 rule ESA ★★★

$$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}, \qquad f(a) \neq 0$$

GIVES The PI of an exponential.

APPLIES $X = e^{ax}$, a positive, negative or zero.

NOTE Replace D by a. For e^{-ax} replace D by $-a$: $\frac{e^{-ax}}{f(-a)}$ (T003 00:10:51).

S003·8 · S003·13 · T003 00:10:15 · T003 00:10:51 · T004a 00:00:41

E3.2 · First case of failure ESA ★★★

$$f(a) = 0: \quad \frac{1}{f(D)}e^{ax} = x\,\frac{e^{ax}}{f'(a)}, \qquad f'(a) \neq 0$$

GIVES The PI when a is a simple root of $f(m) = 0$.

APPLIES $D-a$ is a factor of $f(D)$ once.

NOTE $f'(D) = \frac{d}{dD}f(D)$; x is written outside the operator (T003 00:11:51).

S003·8 · S003·13 · T003 00:11:35 · T003 00:12:08 · T004a 00:01:20 · ESA Jan-24 s2 2d

E3.3 · Second case of failure ESA ★★★

$$f(a) = f'(a) = 0: \quad \frac{1}{f(D)}e^{ax} = x^2\,\frac{e^{ax}}{f''(a)}, \qquad f''(a) \neq 0$$

GIVES The PI when a is a double root.

APPLIES $(D-a)^2$ divides $f(D)$.

NOTE ESA Dec-23 3a: $\frac{8e^{2x}}{(D-2)^2} = \frac{8x^2e^{2x}}{2} = 4x^2e^{2x}$.

S003·8 · S003·13 · T003 00:12:53 · T004a 00:01:41 · ESA Dec-23 3a

E3.4 · Third case of failure ESA ★★

$$f(a) = f'(a) = f''(a) = 0: \quad \frac{1}{f(D)}e^{ax} = x^3\,\frac{e^{ax}}{f'''(a)}$$

GIVES The PI when a is a triple root.

APPLIES $(D-a)^3$ divides $f(D)$; "and so on".

NOTE CW P12: $\frac{4e^{-x}}{(D+1)^3(D-2)} = \frac{4x^3e^{-x}}{f'''(-1)} = -\frac{2}{9}x^3e^{-x}$.

TRANSCRIPT ONLY T003 00:13:19 · S003·13 · CW U2 P12

E3.5 · Root of multiplicity k (shortcut) ESA ★★

$$f(D) = (D-a)^k\varphi(D),\ \varphi(a) \neq 0: \quad \frac{1}{f(D)}e^{ax} = \frac{x^k\,e^{ax}}{k!\,\varphi(a)}$$

GIVES The same answer as the ladder, in one step.

APPLIES The factorised $f(D)$ is known.

NOTE $f^{(k)}(a) = k!\,\varphi(a)$. Jan-24 s2 2d: $\varphi(D) = (D-1)(D-3)$, $\varphi(2) = -1$, PI $= -xe^{2x}$.

ADDED FOR CLARITY ESA Jan-24 s2 2d · ESA Dec-23 3a

4 Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x

E4.1 · Constant right-hand side ESA ★★

$$\frac{k}{f(D)} = k\,\frac{e^{0x}}{f(D)} = \frac{k}{f(0)}, \qquad f(0) \neq 0$$

GIVES The PI of a constant.

APPLIES $X = k$.

NOTE If $f(0) = 0$: $\frac{xk}{f'(0)}$, then $\frac{x^2k}{f''(0)}$, ... (T003 00:14:25). Slide 003·14: $\frac{1}{D^2+2D+5} = \frac{1}{5}$.

S003·9 · S003·14 · T003 00:13:50 · T003 00:14:25 · T004a 00:02:06 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E4.2 · $\sinh ax$ ESA ★★

$$\frac{\sinh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$$

GIVES The PI of $\sinh ax$.

APPLIES $f(a) \neq 0$, $f(-a) \neq 0$; otherwise use the failure ladder on that term.

NOTE $\sinh ax = \frac{e^{ax}-e^{-ax}}{2}$. Slide 003·16: $2\sinh x = e^{x} - e^{-x}$.

S003·9 · T003 00:14:55 · T003 00:15:19 · CW U2 P11

E4.3 · $\cosh ax$ ESA ★★

$$\frac{\cosh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$$

GIVES The PI of $\cosh ax$.

APPLIES $f(\pm a) \neq 0$; otherwise use the ladder.

NOTE $\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$. CW P12: $e^{-x}\cosh x = \frac{1+e^{-2x}}{2}$.

S003·10 · T003 00:15:37 · CW U2 P12

E4.4 · a^{x} and a^{-x} ESA ★★

$$\frac{a^{x}}{f(D)} = \frac{a^{x}}{f(\log a)}, \qquad \frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)}$$

GIVES The PI of a power of a constant.

APPLIES $a > 0$, $f(\pm \log a) \neq 0$.

NOTE $a^{x} = e^{x \log a}$: the exponent is $\log a$ (natural log). ESA Jan-24 s1 2d and s4 2c ask 2^{-x}.

S003·10 · T003 00:15:54 · T003 00:16:34 · T003 00:16:48 · ESA Jan-24 s1 2d · ESA Jan-24 s4 2c

B Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$	all
$f(D)$	Linear differential operator	Polynomial in D with constant coefficients	E1.1–E4.4
$\frac{1}{f(D)}$	Inverse differential operator	Gives the PI: $f(D)\{\frac{1}{f(D)}X\} = X$	E1.1, E1.2
X	Right-hand side	Function of x (capital X, not the variable x)	E1.1–E2.3
y_c, CF	Complementary function	General solution of $f(D)y = 0$	E1.2
y_p, PI	Particular integral	$\frac{1}{f(D)}X$, no arbitrary constants	E1.2
a	Exponent (Type 1)	In e^{ax}; replaces D. Clash: the base in a^x (exponent $\log a$)	E3.1–E3.4, E4.4
$f'(a),\ f''(a)$	Derivatives of f	$\frac{d}{dD}f(D)$, $\frac{d^2}{dD^2}f(D)$ at $D = a$	E3.2–E3.4
k, K	Constant RHS	Also the multiplicity k in E3.5 (added for clarity)	E4.1, E3.5
$\varphi(D)$	Cofactor	$f(D) = (D-a)^k\varphi(D)$, $\varphi(a) \neq 0$ (added for clarity)	E3.5
$\log a$	Natural logarithm	Exponent of $a^x = e^{x\log a}$	E4.4
$\sinh,\ \cosh$	Hyperbolic functions	$\frac{e^{ax} \mp e^{-ax}}{2}$	E4.2, E4.3

Remark. The letter a has two jobs on the slides: the exponent in e^{ax} and the base in a^x. In $a^x = e^{x\log a}$ the exponent is $\log a$, so D is replaced by $\log a$, not by a. X (capital) is the right-hand side, a function of x (small). $\log$ is the natural logarithm. k is a constant right-hand side (the slides write K).